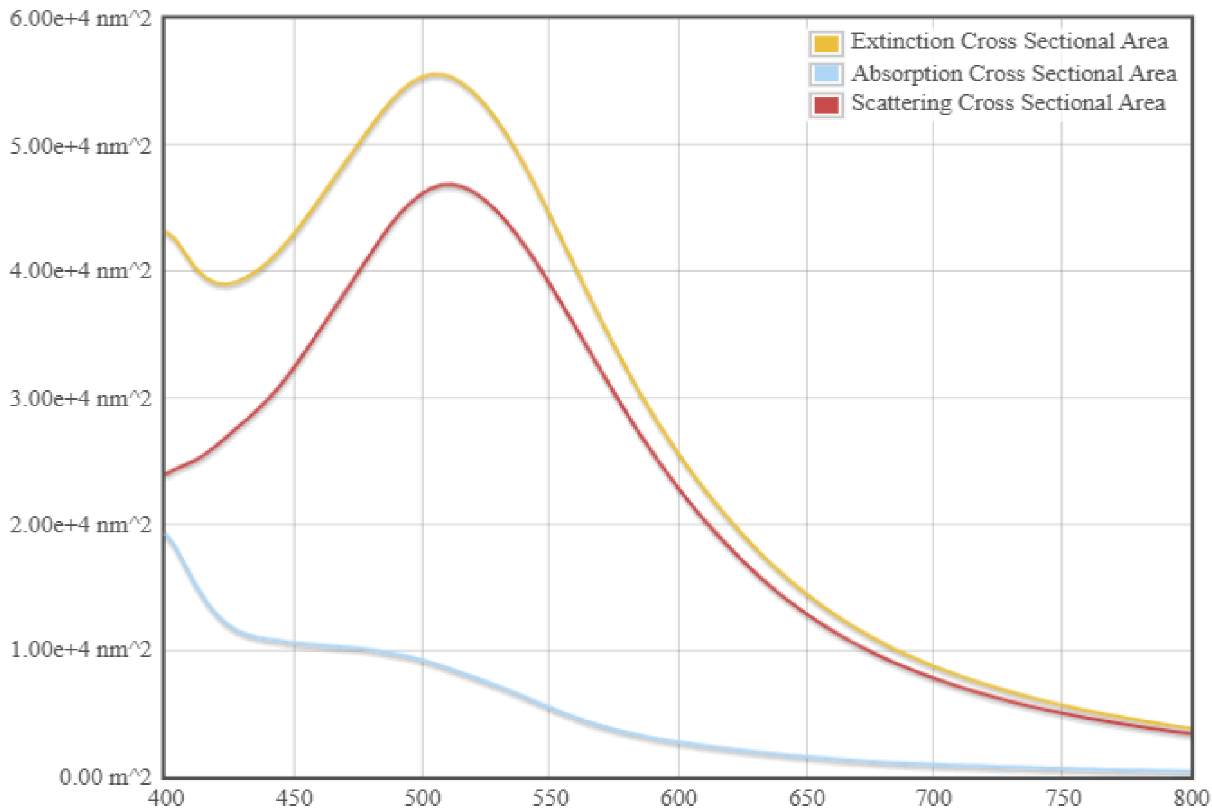
Exp. 1. Simulation of scattering and absorption of Multi-layer Sphere and Core-Shell Nanoparticle using Mie-Plot tool

Scattering and absorption spectra for a core–shell nanoparticle.

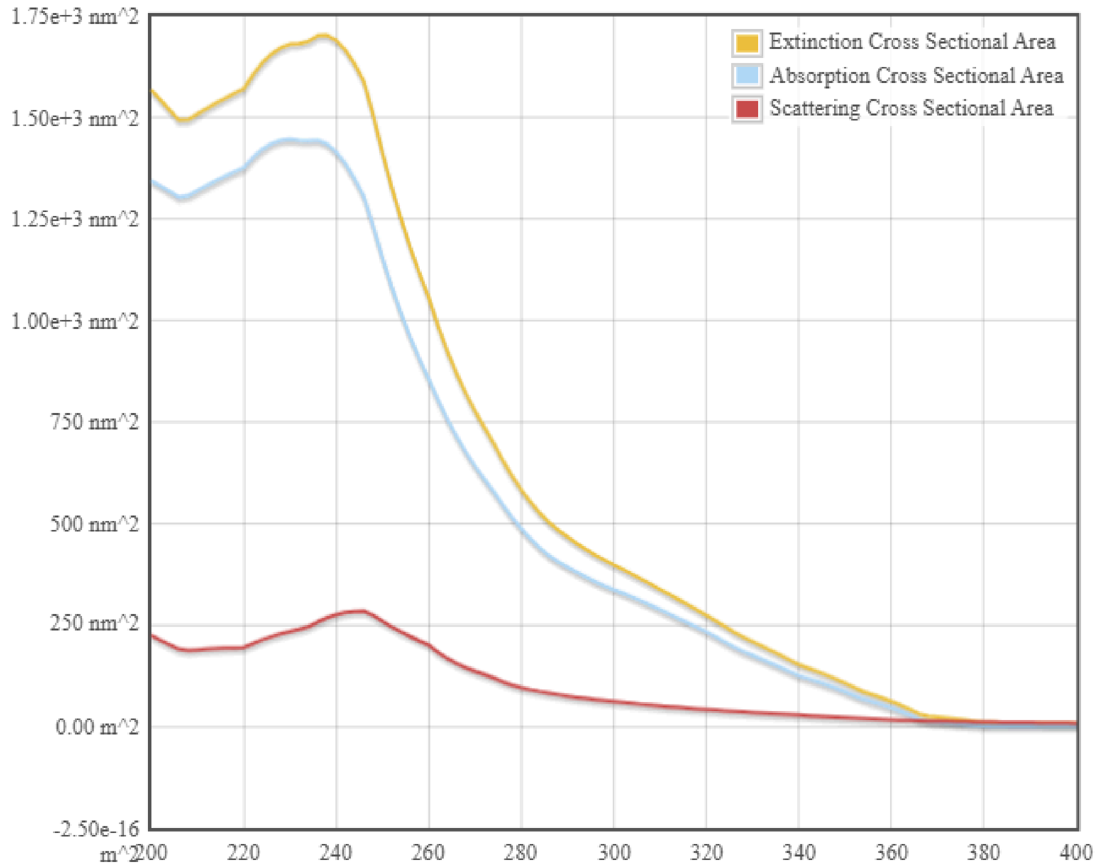


Core/Shell Refractive Indices (RI) only need an input when Core/Shell Type is Other

Warning: your browser is set to not accept cookies. This site is optimized for better performance with cookies; if cookies are not enabled, you will not receive the optimal experience.

Core Type	Core RI	Core Diameter (nm)	
Gold		50	
Shell Type	Shell RI	Shell Thickness (nm)	Plot!
Silver		25	Download Data
Start λ (nm)	End λ (nm)	Medium RI	
400	800	1.33	

Case Study 1: Scattering and absorption spectra for a single-layer sphere (core only).

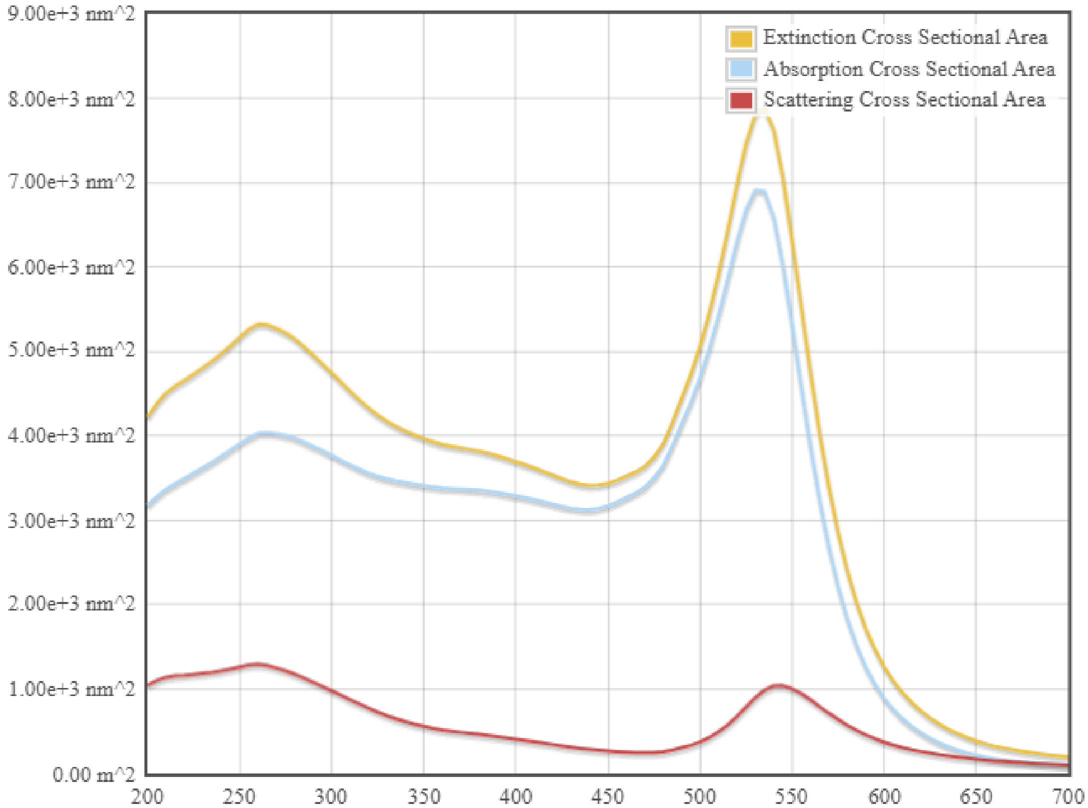


Core/Shell Refractive Indices (RI) only need an input when Core/Shell Type is Other

Warning: your browser is set to not accept cookies. This site is optimized for better performance with cookies; if cookies are not enabled, you will not receive the optimal experience.

Core Type	Core RI	Core Diameter (nm)	
TiO_2		30	
Shell Type	**Shell RI**	**Shell Thickness (nm)**	Plot!
None		0	Download Data
Start λ (nm)	**End λ (nm)**	**Medium RI**	
200	400	1.33	

Case Study 2: Simulate the optical response of a gold (Au) nanoparticle of 50 nm diameter in water medium.

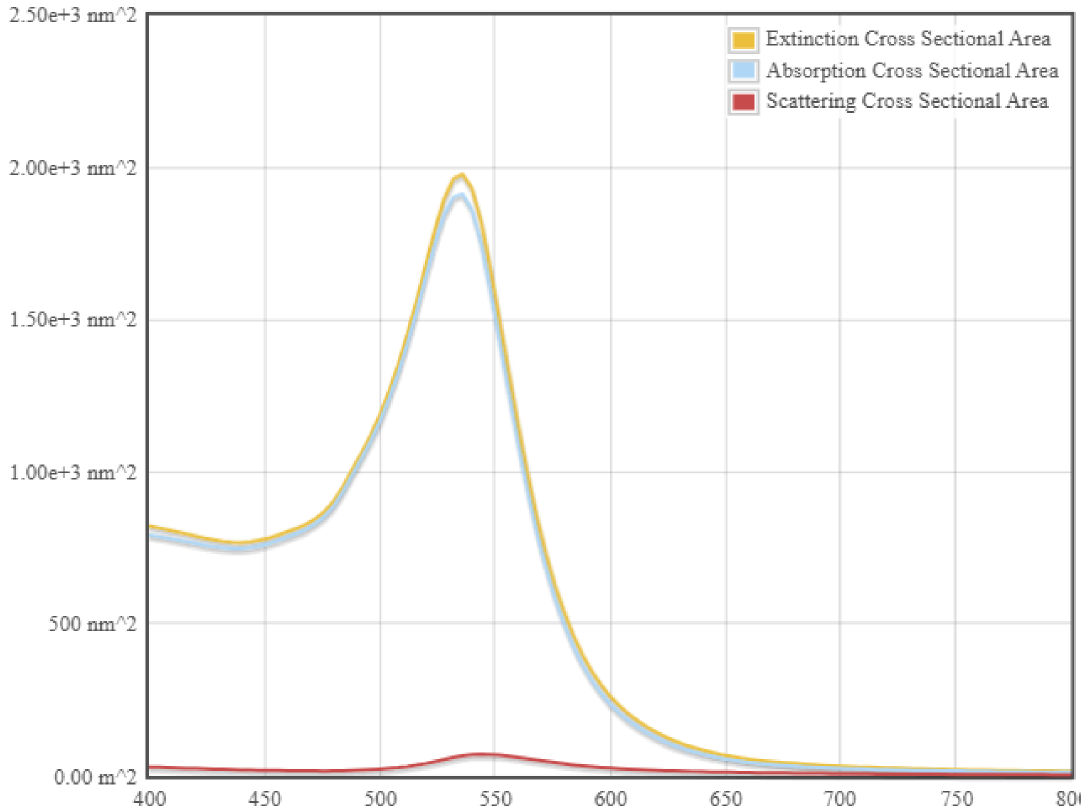


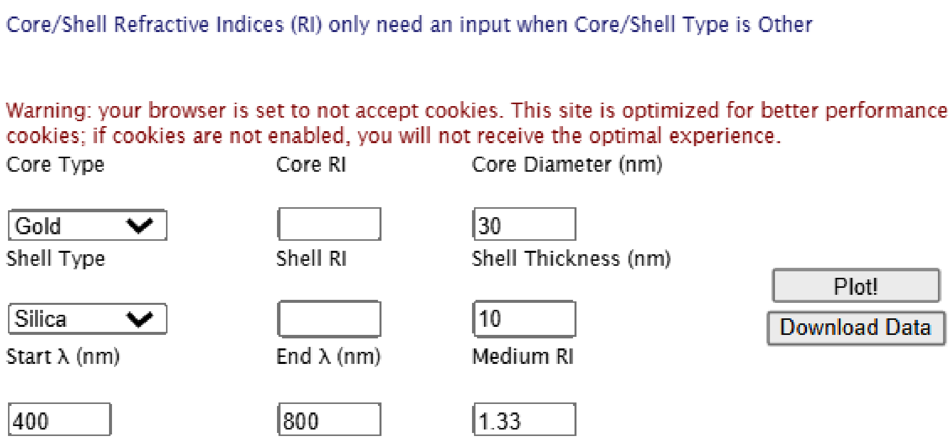
Core/Shell Refractive Indices (RI) only need an input when Core/Shell Type is Other

Warning: your browser is set to not accept cookies. This site is optimized for better performance with cookies; if cookies are not enabled, you will not receive the optimal experience.

Core Type	Core RI	Core Diameter (nm)	
Gold		50	
Shell Type	**Shell RI**	**Shell Thickness (nm)**	Plot!
None		0	Download Data
Start λ (nm)	**End λ (nm)**	**Medium RI**	
200	700	1.33	

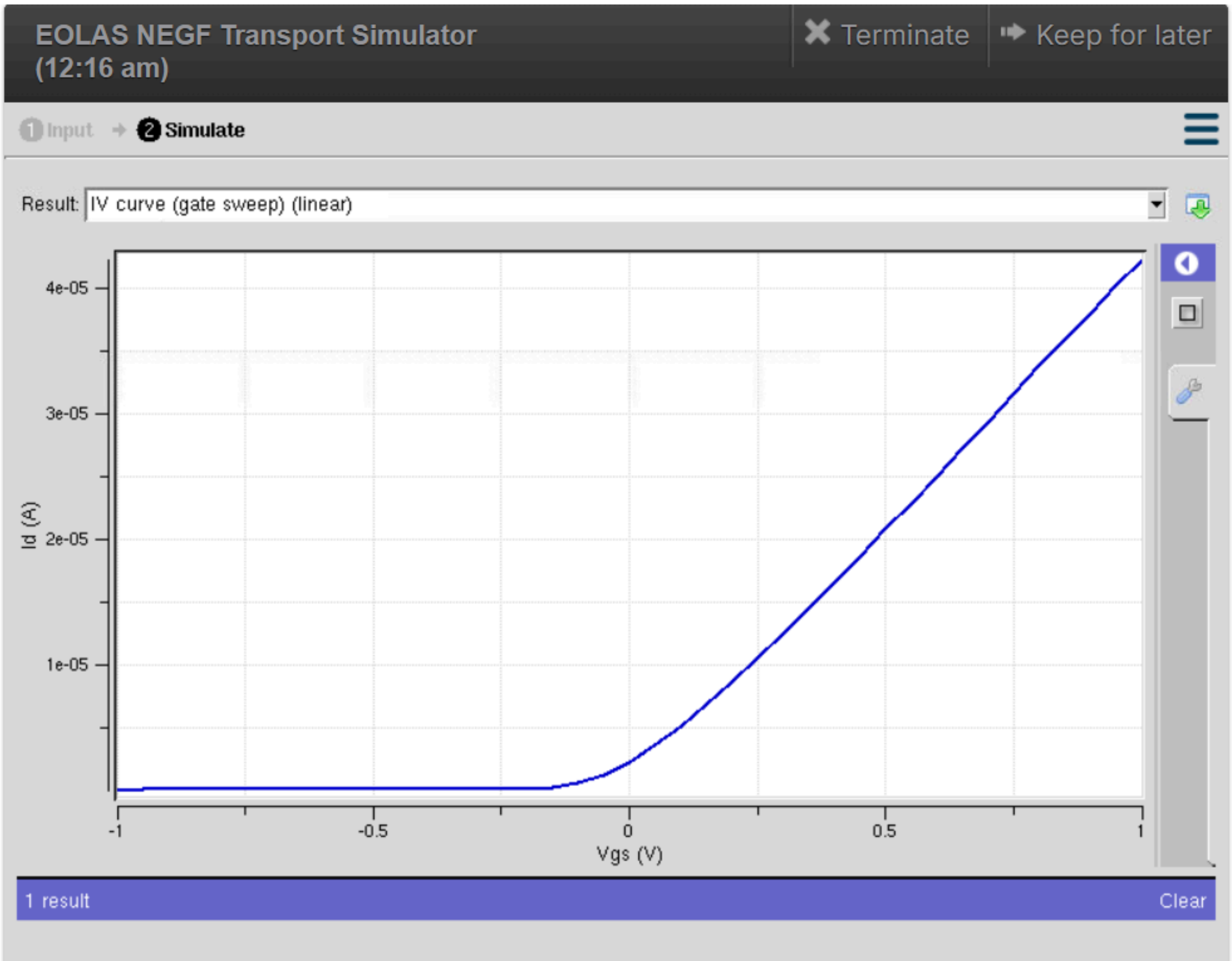
Case Study 3: Simulate a core–shell nanoparticle with a gold core (30 nm) and silica shell (10 nm).





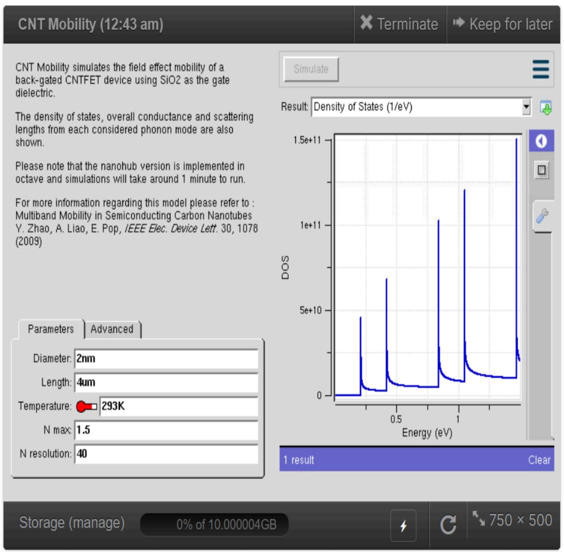
Exp. 8. Simulation of VI Characteristics of FINFET, CNTFET and GFET using NanoHub

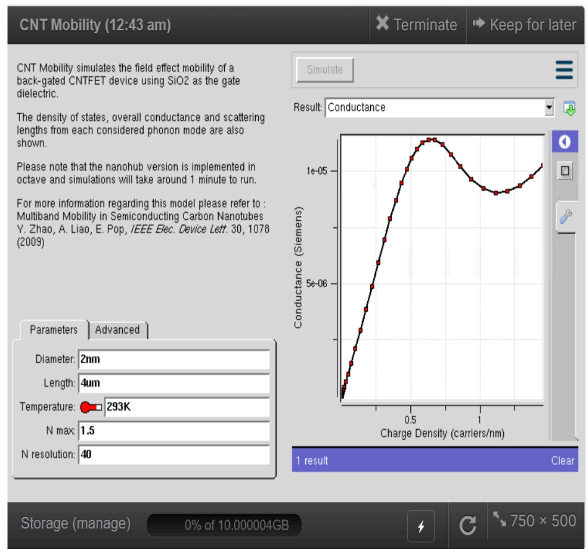
(a) FINFET

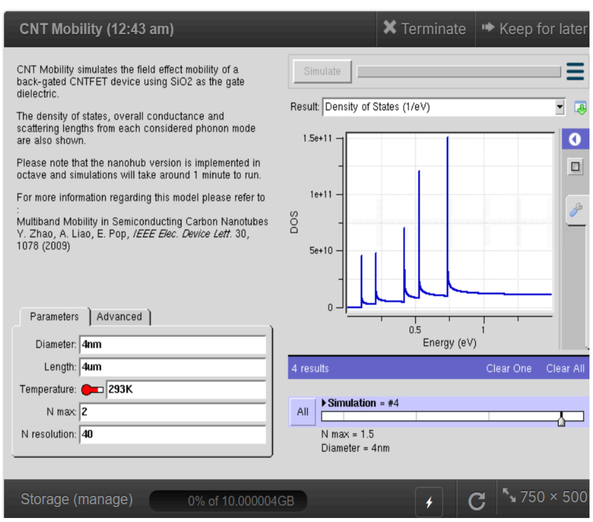


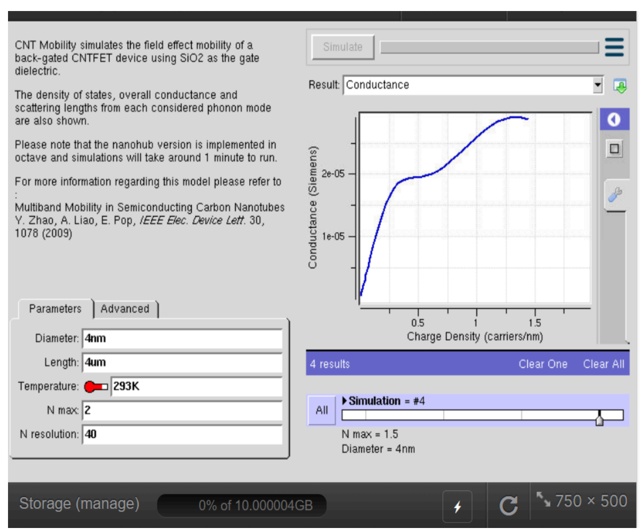
(b) CNTFET

To obtain the density of states and conductance of CNTFET by varying the diameter and length.

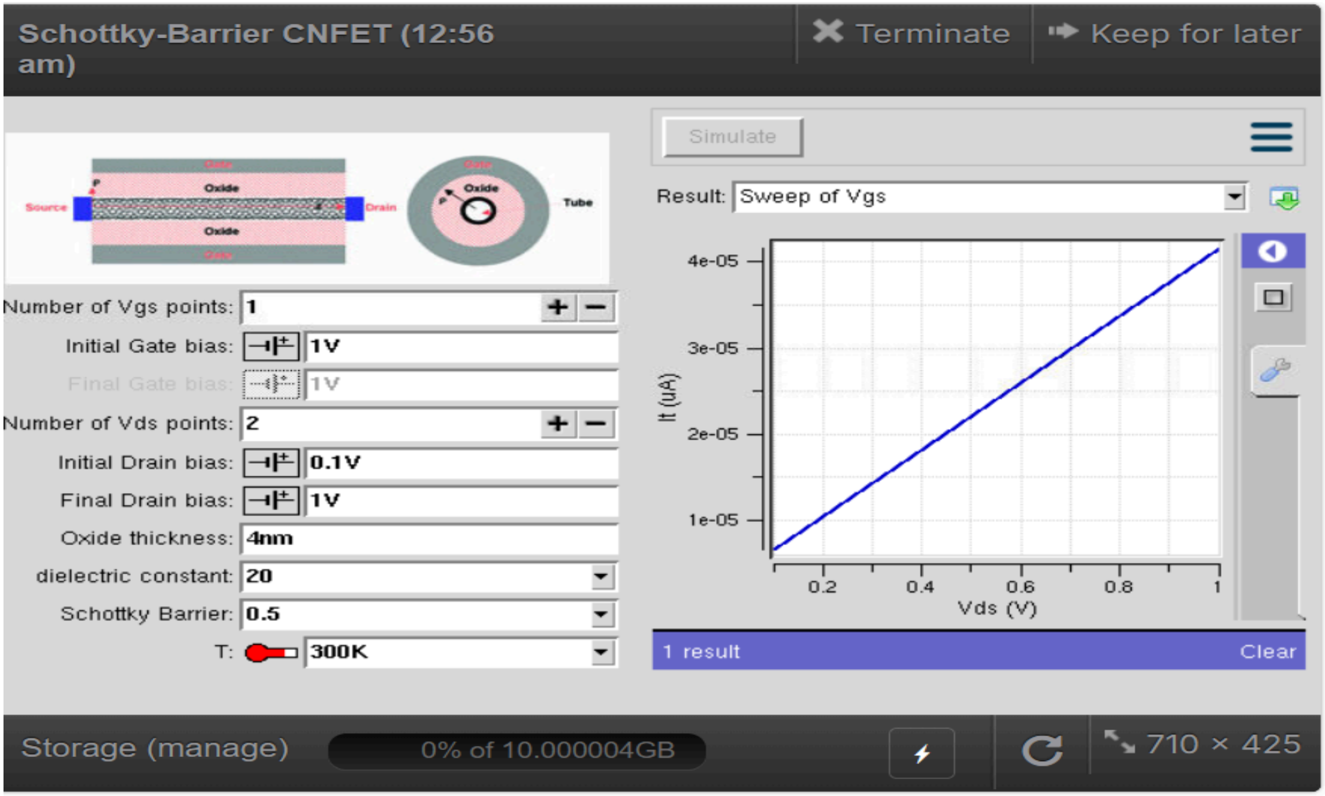




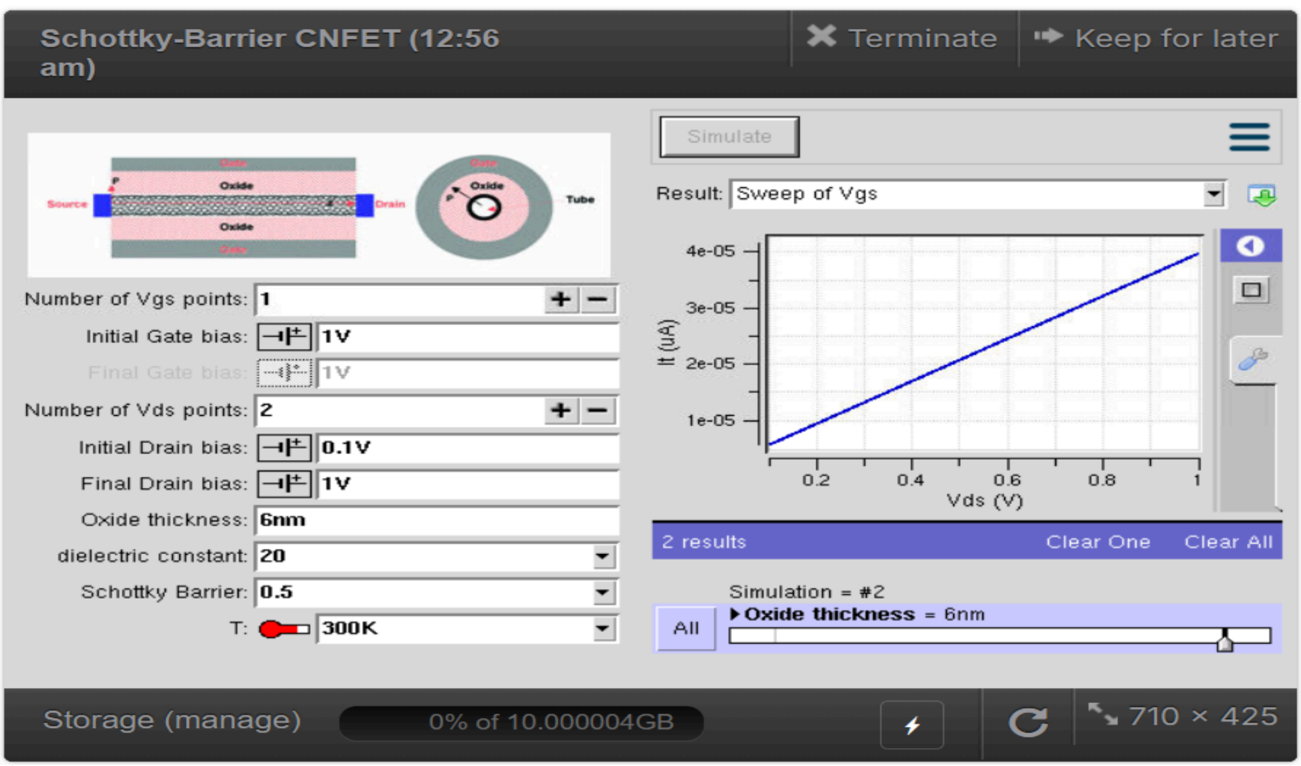




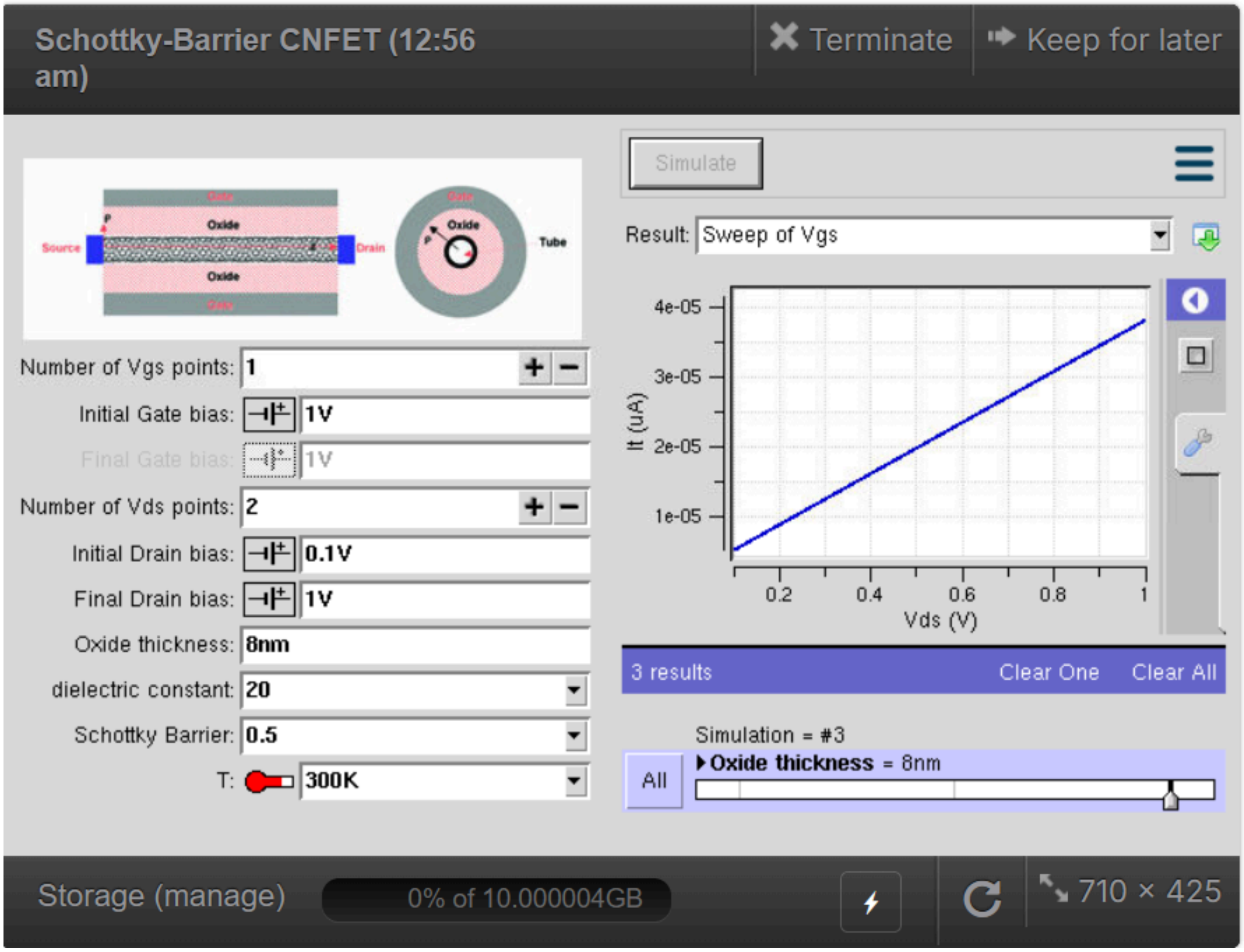
Case study 1: Simulate & Obtain the VI characteristics of CNT-FET by varying Schottky barrier and oxide thickness



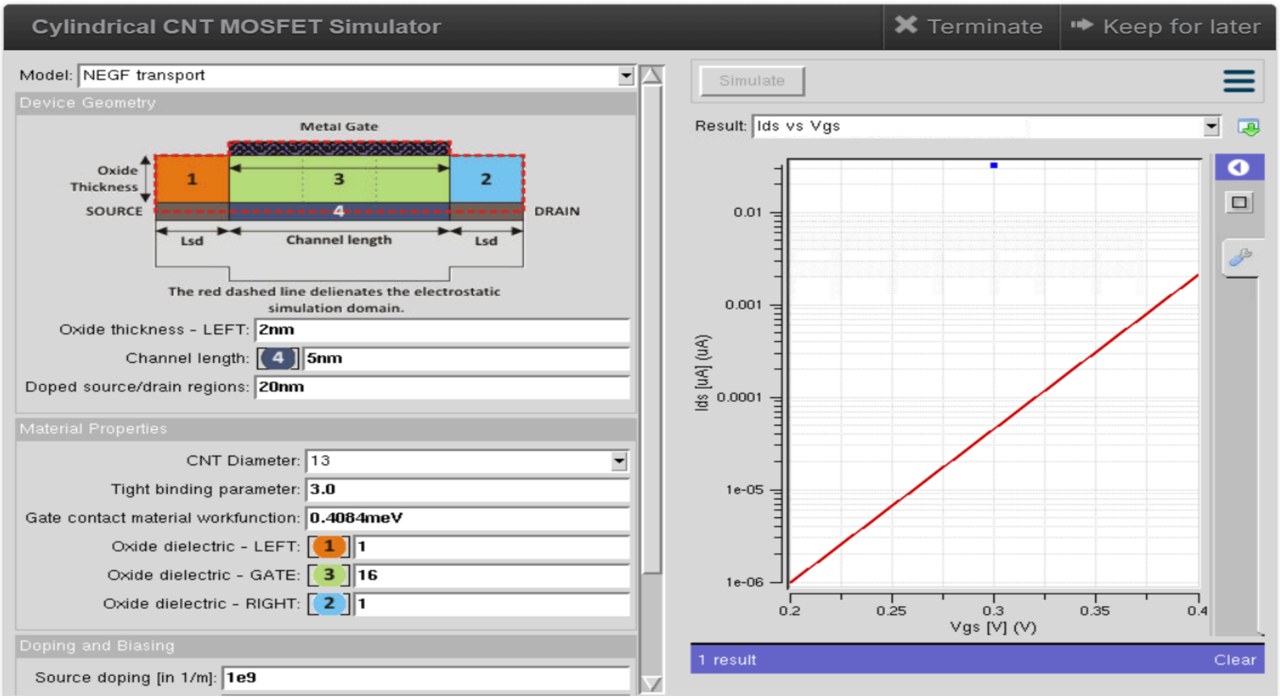
Schottky-Barrier CNFET (12:56 am)
Terminate
Keep for later
Simulate
Result: Sweep of Vgs
Number of Vgs points: 1
Initial Gate bias: 1V
Final Gate bias: 1V
Number of Vds points: 2
Initial Drain bias: 0.1V
Final Drain bias: 1V
Oxide thickness: 4nm
dielectric constant: 20
Schottky Barrier: 0.5
T: 300K
It (uA)
Vds (V)
1 result
Clear
Storage (manage)
0% of 10.000004GB
710 × 425

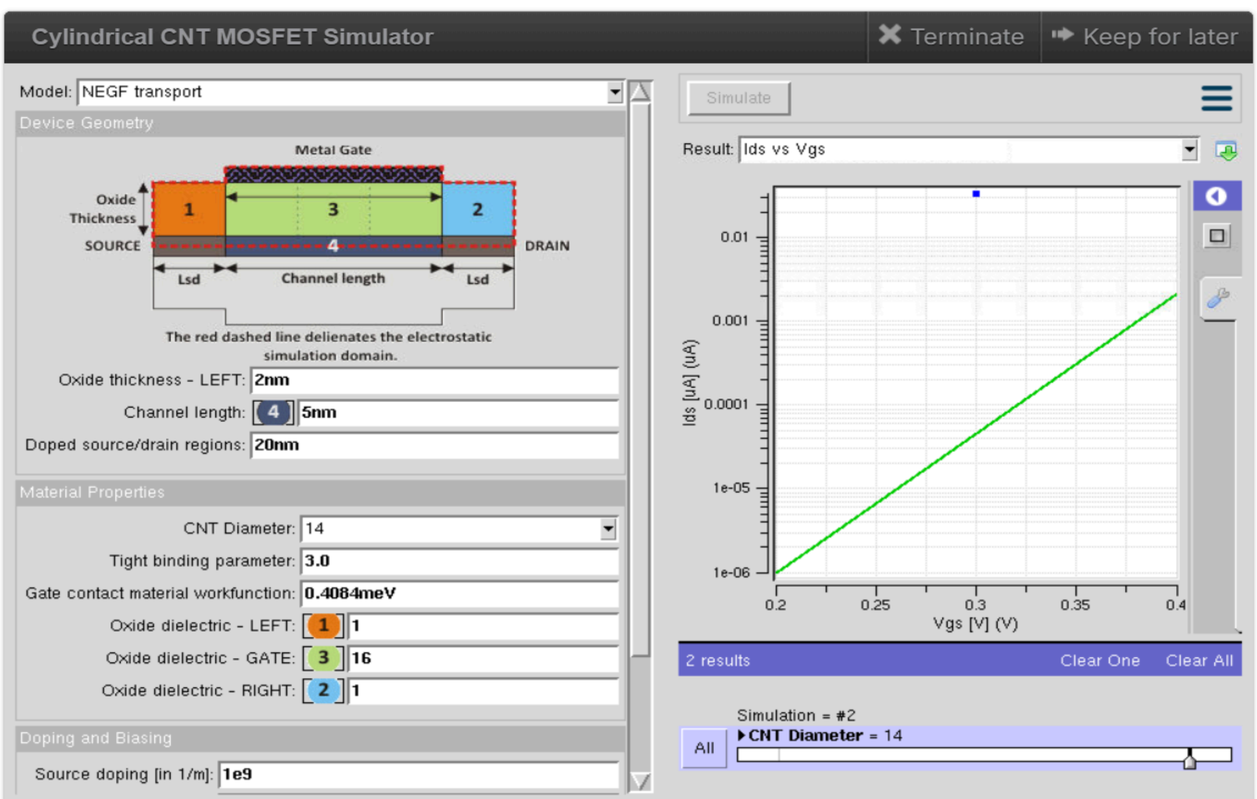


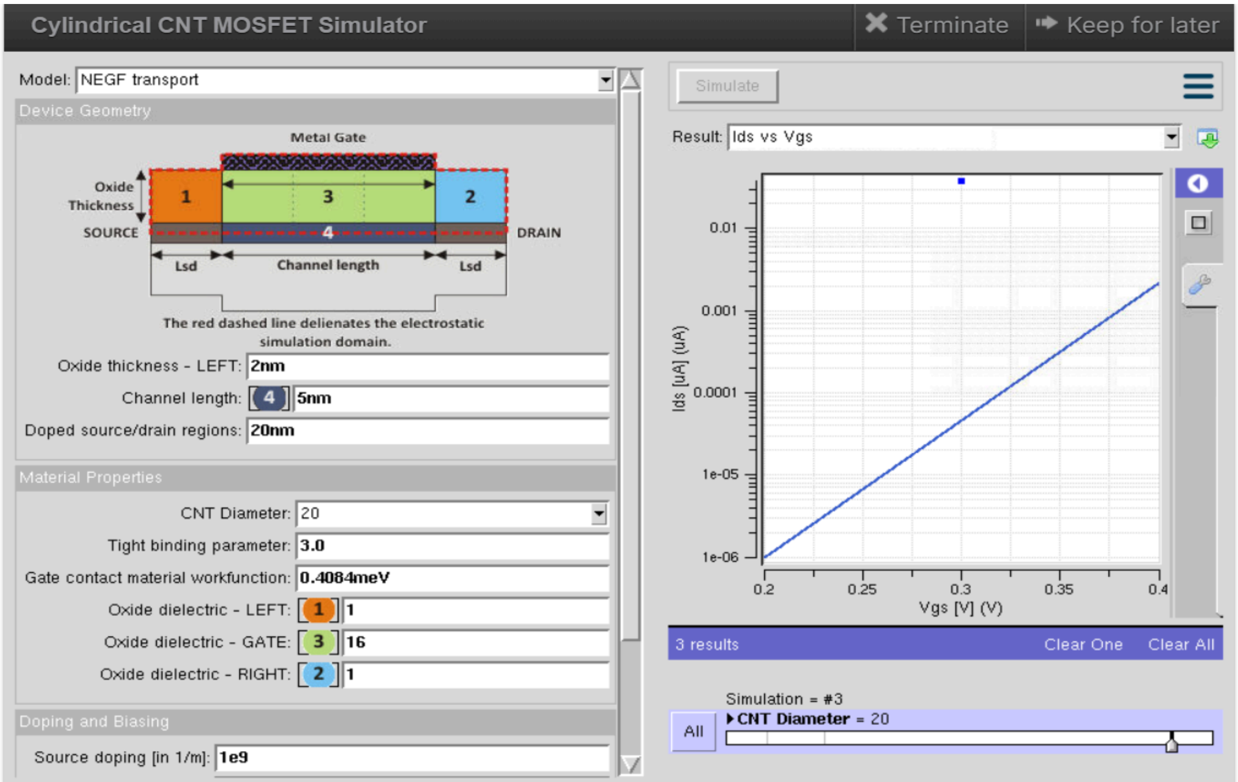
Schottky-Barrier CNFET (12:56 am)
Terminate
Keep for later
Simulate
Result: Sweep of Vgs
Number of Vgs points: 1
Initial Gate bias: 1V
Final Gate bias: 1V
Number of Vds points: 2
Initial Drain bias: 0.1V
Final Drain bias: 1V
Oxide thickness: 6nm
dielectric constant: 20
Schottky Barrier: 0.5
T: 300K
It (uA)
Vds (V)
2 results
Clear One
Clear All
Simulation = #2
Oxide thickness = 6nm
All
Storage (manage)
0% of 10.000004GB
710 × 425


Schottky-Barrier CNFET (12:56 am)
Terminate
Keep for later
Simulate
Result: Sweep of Vgs
Number of Vgs points: 1
Initial Gate bias: 1V
Final Gate bias: 1V
Number of Vds points: 2
Initial Drain bias: 0.1V
Final Drain bias: 1V
Oxide thickness: 8nm
dielectric constant: 20
Schottky Barrier: 0.5
T: 300K
It (uA)
Vds (V)
3 results
Clear One
Clear All
Simulation = #3
Oxide thickness = 8nm
All
Storage (manage)
0% of 10.000004GB
710 × 425

Case study 2: Simulate & Obtain the VI characteristics of CNT-FET by varying CNT diameter



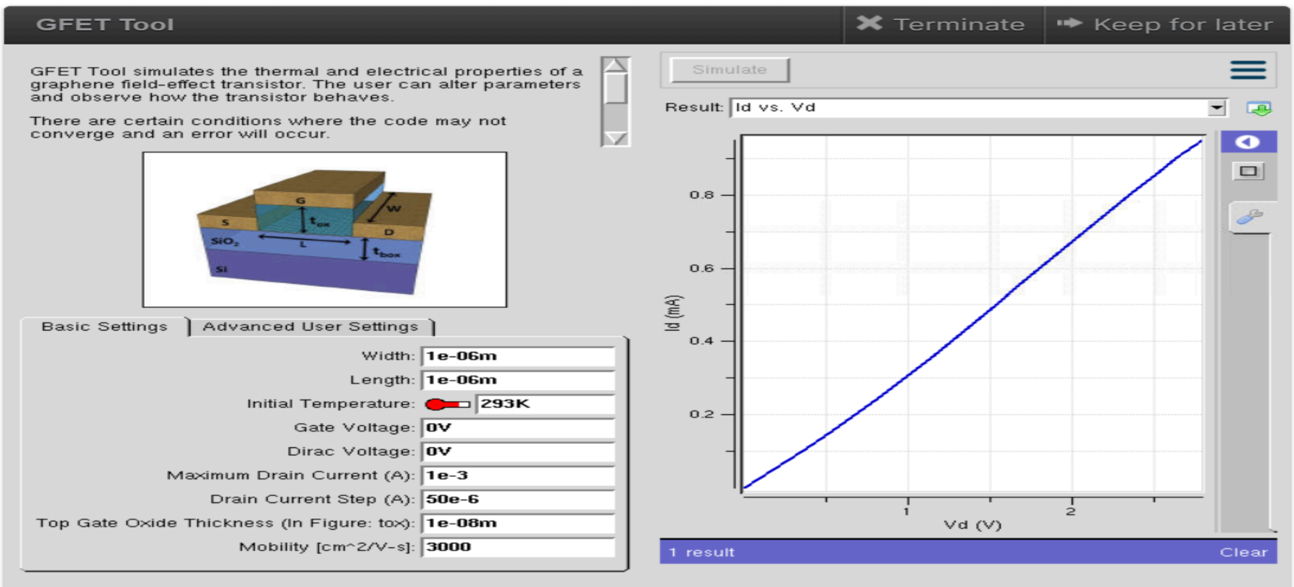




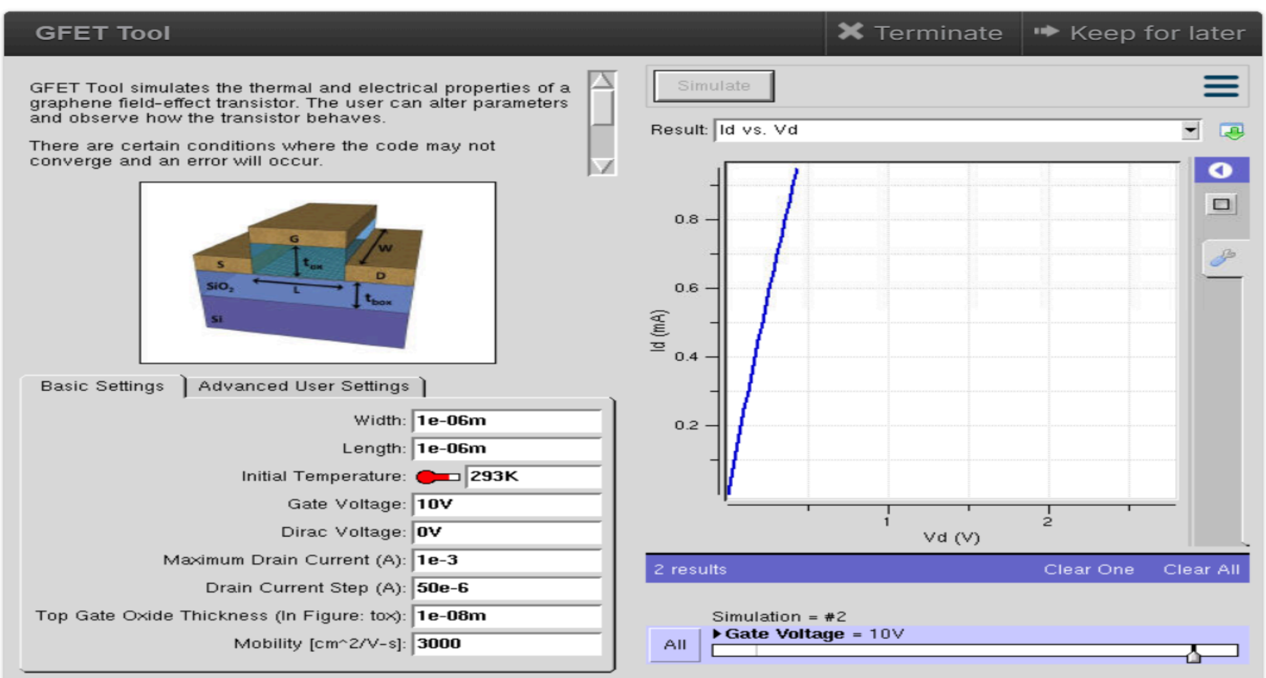
Cylindrical CNT MOSFET Simulator
Terminate
Keep for later
Model: NEGF transport
Device Geometry
Metal Gate
Oxide Thickness
SOURCE
DRAIN
1
3
2
4
Lsd
Channel length
Lsd
The red dashed line delienates the electrostatic simulation domain.
Oxide thickness - LEFT: 2nm
Channel length: 4 5nm
Doped source/drain regions: 20nm
Material Properties
CNT Diameter: 20
Tight binding parameter: 3.0
Gate contact material workfunction: 0.4084meV
Oxide dielectric - LEFT: 1 1
Oxide dielectric - GATE: 3 16
Oxide dielectric - RIGHT: 2 1
Doping and Biasing
Source doping [in 1/m]: 1e9
Simulate
Result: Ids vs Vgs
Ids [uA] (uA)
0.01
0.001
0.0001
1e-05
1e-06
0.2
0.25
0.3
0.35
0.4
Vgs [V] (V)
3 results
Clear One
Clear All
Simulation = #3
All
CNT Diameter = 20

(c) GFET

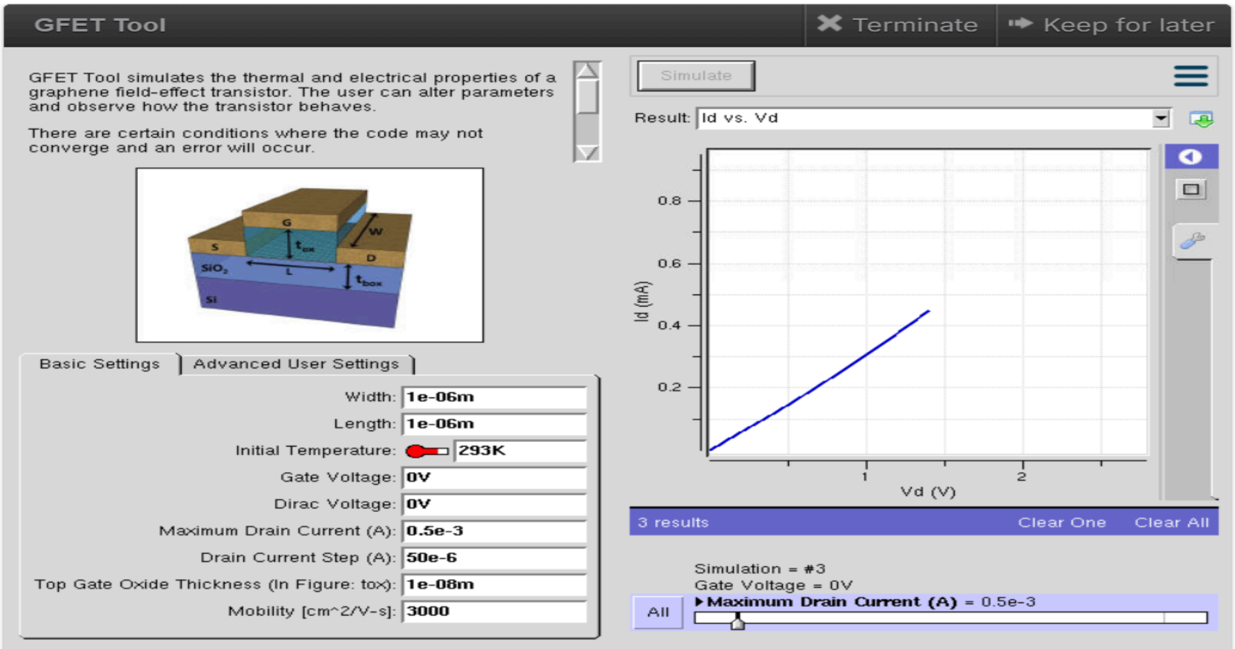
Case Study 1:How does varying the gate voltage (Vgs) affect the drain current (Ids) in a GFET?



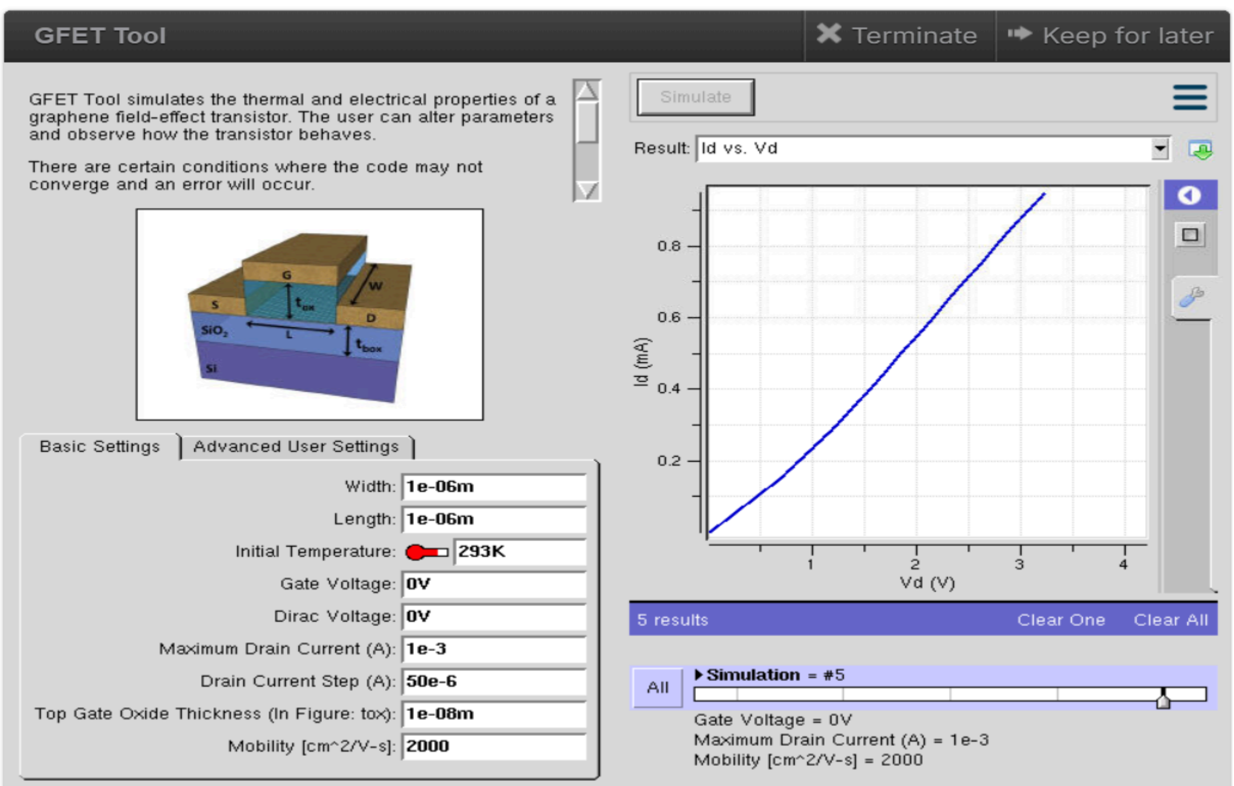
Case Study 1:How does varying the gate voltage (Vgs) affect the drain current (Ids) in a GFET?



Case Study 2: What factors determine the maximum drain current (Ids,max) in a Graphene Field Effect Transistor (GFET)?

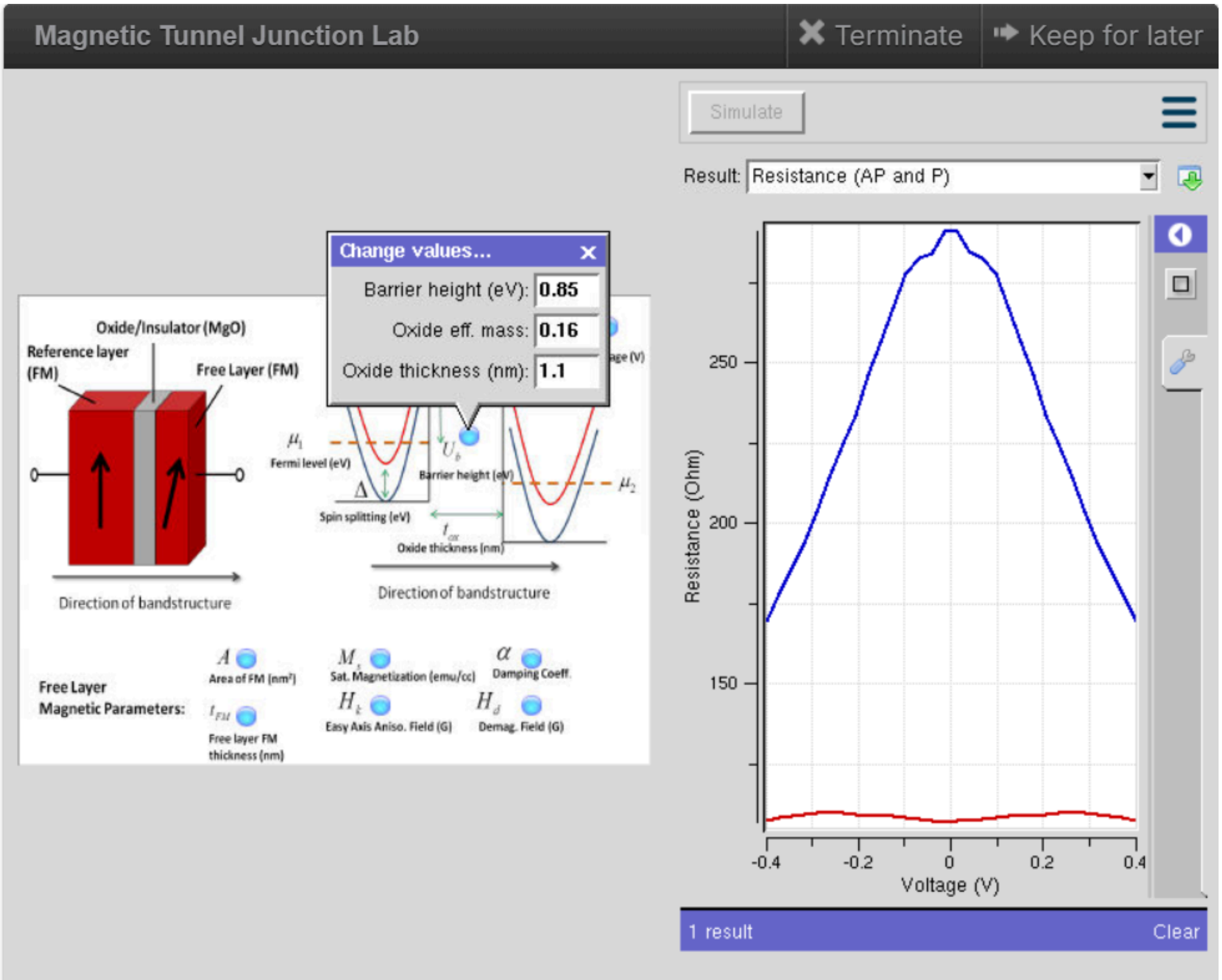


Case Study 3: To investigate how the mobility of charge carriers in a Graphene Field-Effect Transistor (GFET) with a mobility of 2000 cm²/V·s influences its current-voltage (VI) characteristics.



Exp. 9. Simulation of Tunneling Magnetoresistance (TMR) by Varying Barrier Height using NanoHUB

Case Study 1: Characteristics of TMR with barrier height of 0.85eV



Case Study 2: Characteristics of TMR with barrier height of 1.0eV